Query Efficient Structured Matrix Learning

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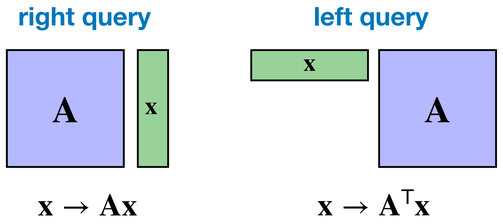
Matrix Learning

Goal: Learn some unknown matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$.

Matrix-vector product access model:

Input: Query vector $\mathbf{x}$

Output: $\mathbf{Ax}$ or $\mathbf{A}^\mathsf{T}\mathbf{x}$



Query Complexity

What's the minimum number of queries required to find the best approximation to matrix $\mathbf{A}$ from a hypothesis class?

Problem: For an unknown target matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, approximation factor $\gamma > 1$, and a matrix family $\mathscr{F} \subset \mathbb{R}^{n\times n}$, find $\tilde{\mathbf{B}} \in \mathscr{F}$ s.t.

$$\|\mathbf{A} - \tilde{\mathbf{B}}\|_F \le \gamma \cdot \min_{\mathbf{B}\in\mathscr{F}} \|\mathbf{A} - \mathbf{B}\|_F$$

Results for specific structured matrices are well studied. The following table shows results for algorithms that achieve $\gamma = (1+\epsilon)$ approximations.

Matrix Family	# of queries
Diagonal	$O(1/\epsilon)$
Rank-k	$O(k/\epsilon^{1/3})$
s-sparse rows	$O(s/\epsilon)$
Rank-k HODLR	$O(k\log^4 n/\epsilon^3)$

Towards a General Theory of Matrix Learning

Does there exist a general theory to characterize the query complexity of structured matrix approximation?

Main Result: Finite Families

Theorem: There is an algorithm that, for any finite matrix family $\mathscr{F} \subset \mathbb{R}^{n\times n}$ and any $\epsilon \in (0,1)$, uses $\tilde{O}\left(\sqrt{\log|\mathscr{F}|}/\epsilon^2\right)$ matrix-vector product queries with $\mathbf{A}$ and $\mathbf{A}^\mathsf{T}$ and finds, with high probability, an approximation $\tilde{\mathbf{B}} \in \mathscr{F}$ satisfying:

$$\|\mathbf{A} - \tilde{\mathbf{B}}\|_F < (3+\epsilon)\cdot \min_{\mathbf{B}\in\mathscr{F}} \|\mathbf{A} - \mathbf{B}\|_F$$

Lower Bound: $\Omega\left(\sqrt{\log|\mathscr{F}|}/\log\log|\mathscr{F}|\right)$. Results are optimal in the worst case.

Infinite Families via Covering Number

Theorem: Consider any $\alpha > 0$. There is an algorithm that, for any matrix family $\mathscr{F} \subset \mathbb{R}^{n\times n}$ with a given α-cover of cardinality Γ_α, and any $\epsilon \in (0,1)$, uses $\tilde{O}\left(\sqrt{\log\Gamma_\alpha}/\epsilon^2\right)$ matrix-vector product queries with $\mathbf{A}$ and $\mathbf{A}^\mathsf{T}$ and finds, with high probability, an approximation $\tilde{\mathbf{B}} \in \mathscr{F}$ satisfying:

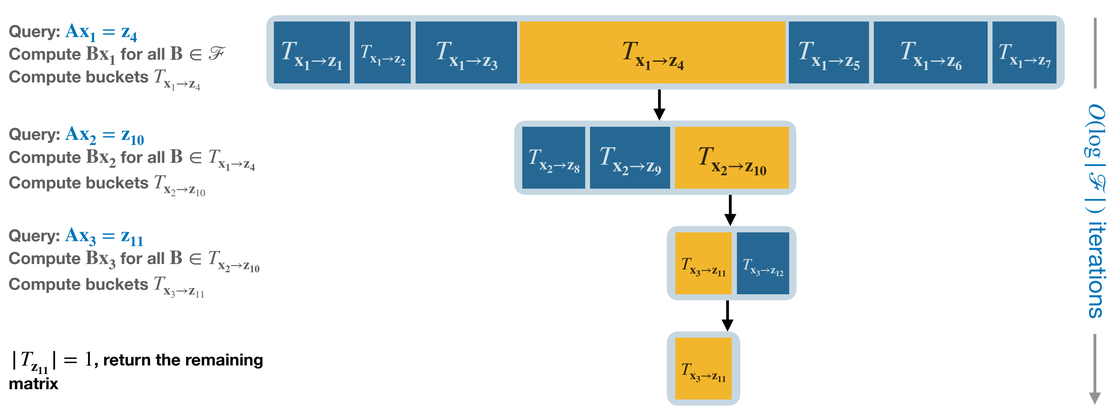
$$\|\mathbf{A} - \tilde{\mathbf{B}}\|_F < (3+\epsilon)\cdot \inf_{\mathbf{B}\in\mathscr{F}} \|\mathbf{A} - \mathbf{B}\|_F + (3+\epsilon)\alpha$$

Realizable Setting: $\mathbf{A} \in \mathscr{F}$

In a simpler variant of this problem, we show that $\tilde{O}(\sqrt{\log|\mathscr{F}|})$ suffice to exactly recover any $\mathbf{A} \in \mathscr{F}$. Algorithm idea extends to agnostic setting.

One-Sided Approach - $O\left(\log|\mathscr{F}|\right)$

Algorithm Idea: Partition the candidate matrices into "buckets" and recurse on the bucket containing $\mathbf{A}$. Let $T_{\mathbf{x}\to\mathbf{z}} = \{\mathbf{B} \in \mathscr{F} : \mathbf{Bx} = \mathbf{z}\}$ be a bucket.



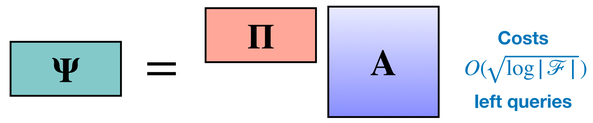
Two-Sided Approach - $\tilde{O}\left(\sqrt{\log|\mathscr{F}|}\right)$

Problem with one-sided algorithm:

"Large" buckets $\Longrightarrow$ more iterations

Idea: Use a sketch of $\mathbf{A}$ to avoid large buckets.

Sketching matrix: $\mathbf{\Pi} \in \{\pm 1\}^{O(\sqrt{\log|\mathscr{F}|})\times n}$



Say, $T_\mathbf{z}$ is "**large**" if $|T_\mathbf{z}| > |\mathscr{F}|/2^{\sqrt{\log|\mathscr{F}|}}$ and let $\mathbf{Ax} = \mathbf{z}^*$.

Case 1: $\mathbf{A}$ is in large bucket

- Show there's a unique large $T_{\mathbf{x}_1\to\mathbf{z}}$ s.t. $\mathbf{\Pi A x}_1 = \mathbf{\Pi z}$.
- **No additional queries needed**

Case 2: $\mathbf{A}$ is in small bucket

- Show that $\mathbf{\Pi A x}_1 \neq \mathbf{\Pi z}$ for any large $T_{\mathbf{x}_1\to\mathbf{z}}$
- **Need 1 right query to get $\mathbf{Ax} = \mathbf{z}^*$**

Case 1 happens at most $\log|\mathscr{F}|$ times.

Case 2 happens at most $\sqrt{\log|\mathscr{F}|}$ times.

Matrix Recovery Applications

- Linear PDE learning (operator learning)
- Structured solver compression
- Low-rank approximation
- Learning the Hessian of some optimization problem.

Future Work

- Another measure of complexity that can achieve instance optimality
- $(1+\epsilon)$-approximation algorithm
- Computationally efficient algorithms for families with compact representations, e.g., linear matrix families